MIST 7770: Business Intelligence & Analytics

Spring 2026

Monday and Wednesday 1:15 pm – 2:35 pm, Correll Hall 0315 (CRN: 67560)

Instructor Information

Name: Bedile Kara
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Office: C404 Benson Hall
Office Hours: Online/Face-to-Face Monday/Wednesday 3:30 pm to 4:30 pm and **by appointment**. The appointments for the Face-to-Face office meetings should be scheduled by the end of the class. Find zoom meeting information on the last page of the syllabus and eLC.
Contact Notes: Office Hours are on a first-come, first-served basis.

Course Delivery
This course meets in person (times and days indicated above).

Course Description
This course provides a broad understanding of technologies and processes for gathering, storing, accessing, and analyzing data to help business users make better decisions. Topics include problem formulation and research design; data collection, manipulation, visualization, and modeling; big data, machine learning (ML), and storytelling with data; robots, automation, and the promise of artificial intelligence. This course includes the use of generative AI and trains students on prompt engineering and the capabilities of generative AI in business intelligence. These topics are divided into four sections:

(1) business analytical framework,
(2) data gathering and pre-processing,
(3) data analysis and presentation,
(4) next generation business intelligence.

Please note that the instructional approach may change as need warrants.

Required pre-requisites
Permission of the department.

Changes to the Syllabus/Course
The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.

Updates, changes or clarifications to the syllabus, assignments, or other course-related matters will typically be addressed at the beginning or end of class. I also occasionally post updates to the eLC website, send notification via eLC's email function, or to your official UGA email address. You are responsible for updates via any of these mechanisms.

Objectives

Students completing this course will be able to:

- Develop research design based on the formulation of a business problem
- Collect, manipulate, visualize, and model data in Tableau, R, and Gephi
- Learn the basics of big data and machine learning for business
- Develop expertise in prompt engineering with generative AI
- Apply generative AI to real-world business intelligence and analytics scenarios
- Leverage storytelling for presenting data insights
- Understand future trends and directions in business intelligence

Textbooks

You can access all these books for free in the links below:

-Wickham, H., & Grolemund, G. (2017). R for Data Science: Import, tidy, transform, visualize, and model data (eBook in authors' website)

-Provost, F. & Fawcett, T. (2013). Data Science for Business (eBook in UGA Library)

-Knaflic, C. G. (2015). Storytelling with Data (eBook in UGA Library)

Software

R, RStudio, and Tableau for data manipulation, visualization, and modeling. Choose the **open-source desktop** version of RStudio. Request a Free Student License in Tableau for one year (use *uga.edu* e-mail). Gephi for social network analysis. ChatGPT for generative AI. Sign up for a free ChatGPT account (use *uga.edu* e-mail).

General course policies

- Attendance is not required for this course but highly encouraged. A recent PNAS study shows that "*being actively engaged in the classroom*" is a significant predictor of actual learning.
- Make-up exams may only be given for valid and documented reasons, at the discretion of the instructor. An unexcused absence from an exam will lead to zero points earned on that exam.
- Backup Policy: Always keep a copy of all material turned in for grading.
- This course teaches the use of artificial intelligence software to craft effective prompts to optimize coding. Students will learn how to guide generative AI output and develop code through generative AI to apply business intelligence and analytics models to real-world scenarios.
- Students are required to disclose the use of generative AI software in any assignments and must submit the prompts used to generate their outputs.
- Students are responsible for the quality and accuracy of the work submitted, regardless of the use of generative AI.
- All discussions related to grades will be conducted via e-mail only.

Class Format

This class uses **flipped classroom** and **cooperative methods** of learning. In a **flipped classroom**, students take responsibility for learning materials before attending class (via eLC). Class time is spent reviewing the understanding and application of knowledge through discussion questions and exercises. In **cooperative learning**, small teams comprising participants with varying levels of ability learn together. Each team member is responsible for his or her own learning as well as

for helping the team learn. Cooperative methods of learning are designed to ensure an atmosphere of collective achievement.

File Security
You are responsible for the security of your own files. To avoid problems, I recommend regularly backing up copies of your work. In addition, when modeling or visualizing data in R and Tableau, save your files to prevent data loss in the case of a computer crash.

Academic Honesty
As a University of Georgia student, you have agreed to abide by the University's academic honesty policy, "A Culture of Honesty, " and the Student Honor Code. *UGA Student Honor Code: "I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others." A Culture of Honesty, the University's policy and procedures for handling cases of suspected dishonesty, can be found at http://www.uga.edu/ovpi*. Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation. Questions related to course assignments and the academic honesty policy should be directed to the instructor.

Teamwork
In this course, you will work in groups. As a result, review a short report on team effectiveness and establish a team agreement (sample agreement) for use by the team. It occasionally happens that someone in a team is not prepared to do his/her share. In my courses, I recommend that the team gives the freeloader one warning and then fires that person. That person will then do group assignments individually or find a new team to join. The instructor should be notified of changes in team composition immediately. **All teams should have about 3-5 members**.

Special Accommodations
For special accommodations, especially for quizzes, exams, or assignments, inform the instructor ahead of time (at least 48 hours a priori).

UGA Well-being Resources
UGA Well-being Resources promote student success by cultivating a culture that supports a more active, healthy, and engaged student community. Anyone needing assistance is encouraged to contact Student Care & Outreach (SCO) in the Division of Student Affairs at 706-542-8479. Student Care & Outreach helps students navigate difficult circumstances by connecting them with the most appropriate resources or services. They also administer the Embark@UGA program which supports students experiencing, or who have experienced, homelessness, foster care, or housing insecurity. UGA provides both clinical and non-clinical options to support student well-being and mental health, any time, any place. Whether on campus, or studying from home or abroad, UGA Well-being Resources are here to help.

- Well-being Resources,
- Student Care and Outreach,
- University Health Center,
- Counseling and Psychiatric Services, CAPS 24/7 crisis support at 706-542-2273,
- Health Promotion/ Fontaine Center,
- Disability Resource Center and Testing Services.

Additional information, including free digital well-being resources, can be accessed through the UGA app or by visiting https://well-being.uga.edu.

Diversity and Inclusive Excellence
The Terry College of Business is committed to promoting an inclusive learning and working environment among its students, faculty, and staff. This class welcomes the open exchange of ideas and values freedom of thought and expression and provides a professional environment that recognizes the inherent worth of every person. It aims to foster dignity, understanding, and mutual respect among all individuals in the class.

Assignments
See schedule for the due dates. The due time is **11:59pm (EST)** on the due date. Submit all assignments on eLc (***not e-mail***). I will deduct points for submitting assignments late (10 percent of total points for each day, up to 2 days). Assignments submitted after 2 days of the due date and time will not be accepted.

The purpose of the assignments related to online courses is to provide you with detailed external knowledge and skills. Each of these online courses will offer you a certificate, which you can list in your CV and *make your profile more attractive to employers*.

ID	Topic	Description	Points
a1	Visualization	Complete Udemy's Tableau module	5
a2	Using R	Complete DataCamp's Introduction to R module	5
a3	Modeling	Complete DataCamp's Introduction to Tidyverse	5
a4	Modeling	Complete DataCamp's Statistics module	5
a5	Machine learning	Complete DataCamp's Machine Learning module	5
a6	Analytics	Complete DataCamp's Human Resources Analytics: Exploring Employee Data in R module	5

You will need to pay to complete Udemy's module (**a1**). Make sure to look out for discounts, as of now (Jan 1) the price is **$14.00 per month**. After completion, submit a pdf file of your certificate (on eLc). In addition, write a paragraph including the following:

- Description of the content that you learnt.
- Description of your learning experience in the platform.
- Description of the practical applications or potential use of what you learnt in the course.

To complete the required DataCamp modules (**a2, a3, a4, a5, and a6**) you will need to:

- Sign up for an account using your full name and UGA e-mail address
- Join the **Business Intelligence and Analytics** team (an invite link is available in eLC announcements section)
- When completing a module, submit (on eLc) a pdf file of your certificate. Similar to Udemy, write a paragraph about your experience.

Participation
Once in class, you are expected to contribute to class discussions and exercises as well as clarify your doubts. You are expected to observe respectful behavior, such as not interrupting other students, not using electronic devices unless required for problem solving exercises, and not causing any distractions to fellow students.

TechTalks
One "TechTalk" will be scheduled in most seminars, starting from class 3. Each TechTalk duration is 3-4 minutes only. The "Tech Talks" topic in a class is the previous class' topic or unless otherwise announced. The detailed guidelines for TechTalks are available in eLC.

Quiz
One **short quiz** worth 5 points, with multiple choice and True/False questions will be given during the class.

Group project
Define a problem, use R, Tableau, and/or Gephi to explore data related to the problem, and prepare a presentation for the class of your analysis along with related recommendations. The specific project guidelines will be available on eLC.

Final exam
Final exam will be cumulative. It will be based on UGA schedule of exams:
https://reg.uga.edu/general-information/calendars/final-exam-schedule/

- Correll Hall 0315 (CRN: 67560): Wednesday, April 29th , 12:00 - 3:00 pm

Grading

Item	**Points**
Assignments	30
Participation in discussions and in-class exercises	10
TechTalks	5
Quiz	5
Group project -Project report (Group) 20% -Presentation (Individual) 10%	30
Final exam	20
Total	*100*

Course grades are assigned using the following ranges:

- A = 93 or above
- A- = 90 to 92.99
- B+ = 87 to 89.99
- B = 83 to 86.99
- B- = 80 to 82.99
- C+ = 77 to 79.99
- C = 73 to 76.99
- C- = 70 to 72.99
- D = 60 to 69.99
- F = 59.99 or below

Grades will not be rounded up, for example, an 89.99 is a B+ and will not be changed to a 90.

Schedule

Class	Day	Date	Topic	Assignment	TechTalks
1	M	12-Jan	Syllabus		
2	W	14-Jan	Problem Formulation		
3	W	21-Jan	Research Design		Yes
4	M	26-Jan	Data Collection		Yes
5	W	28-Jan	Data Visualization I	a1	Yes
6	M	2-Feb	Data Visualization II		Yes
7	W	4-Feb	Data Manipulation I	a2	Yes
8	M	9-Feb	Data Manipulation II	a3	Yes
9	W	11-Feb	Data Manipulation III		Yes
10	M	16-Feb	Generative AI and Prompt Engineering	a4	Yes
11	W	18-Feb	Model Basics		Yes
12	M	23-Feb	Project Day 1		Yes
13	W	25-Feb	Simple Linear Regression	Quiz	
14	M	2-Mar	Multiple Linear Regression		Yes
15	W	4-Mar	Logistic Regression		Yes
16	M	16-Mar	Modeling Exercises and Generative AI		Yes
17	W	18-Mar	Project Day 2	a5	Yes
18	M	23-Mar	Social Network Analysis I		Yes
19	W	25-Mar	Social Network Analysis II		Yes
20	M	30-Mar	Big Data & Machine Learning		Yes
21	W	1-Apr	Supervised Learning and Unsupervised Learning	a6	Yes
22	M	6-Apr	Data Storytelling I and Data Storytelling II		Yes
23	W	8-Apr	Data Storytelling III		Yes
24	M	13-Apr	Project day 3		Yes
25	W	15-Apr	Project day 4		Yes
26	M	20-Apr	TBA		
27	W	22-Apr	Project Presentations		
28	M	27-Apr	Course Review		

Note: One or two "TechTalks" will be scheduled in most seminars, starting from class

Zoom Meeting Information- Office Hours

Bedile Kara is inviting you to a scheduled Zoom meeting.
Join Zoom Meeting
https://zoom.us/j/7851490651

Meeting ID: 785 149 0651

One tap mobile
+16469313860,,7851490651# US
+13017158592,,7851490651# US (Washington DC)